

Development Automation **& Linux Lab**

Experiment 4

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EXPERIMENT-4

Functions in Shell Scripting

1. Write a function to fetch two command line arguments (Integers) and then add them and display the result.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ cat ques1.sh
#!/bin/bash

add(){
    echo $(( $1+$2 ))
}

add $1 $2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ ./ques1.sh 7 4
11
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

2. Write a function to input a three-digit number from the user and check if the number is even or odd. If the number is even then return the sum of the digits of the number otherwise return the product of the given number.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ cat ques2.sh
#!/bin/bash

sumofdig(){
    num=$1
    sum=0
    while [ $num -gt 0 ]
    do
        sum=$((sum+((num%10))))
        num=$((num/10))
    done
    echo "Sum of digits: $sum"
}

proofdig(){
    num=$1
    pro=1
    while [ $num -gt 0 ]
    do
        pro=$((pro*$(num%10)))
        num=$((num/10))
    done
    echo "Product of digits: $pro"
}

checkk(){
    num=$1
    if [ $((num%2)) -eq 0 ]; then
        sumofdig $num
    else
        proofdig $num
    fi
}

checkk $1
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ ./ques2.sh 123
Product of digits: 6
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ ./ques2.sh 124
Sum of digits: 7
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

3. Write a function to compute the factorial of a given number N which is input by the user from the terminal.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ cat ques3.sh
#!/bin/bash

fact(){
    f=1
    for (( i=1; i<=$1 ; i++ ));
    do
        f=$((f*i))
    done
    echo $f
}

fact $1
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ ./ques3.sh 7
5040
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

4. Write a function to read the number of lines of a given text file (starting from line 1) as specified by the user. For example, if there is a file demo.txt, and the user specifies that he needs the first 7 lines of the text then display the first 7 lines of demo.txt. Similarly, if the user specifies the last 10 lines, then the last 10 lines of demo.txt should be printed on the terminal.

Code:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ cat ques4.sh
#!/bin/bash

lines(){
    if [[ $from == "S" || $from == "s" ]]; then
        echo $(head -n $2 $1)
    elif [[ $from == "E" || $from == "e" ]]; then
        echo $(tail -n $2 $1)
    else
        echo "Wrong Choice"
    fi
}

read -p "Enter File Name: " filename
read -p "Enter Number of Lines: " lines
read -p "Enter S for Start & E for End: " from
echo
lines $filename $lines $from
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$ ./ques4.sh
Enter File Name: test.txt
Enter Number of Lines: 5
Enter S for Start & E for End: S

Language is one of the most powerful tools humans possess. It allows us to communicate our thoughts, feelings, and ideas to others, and to understand and appreciate the thoughts, feelings, and ideas of others. Every language has its own unique vocabulary, grammar, and syntax, which reflect the culture and history of the people who speak it. Learning a new language opens up new opportunities and experiences, as well as improving cognitive abilities and enhancing cultural understanding. One of the most obvious benefits of learning a new language is the ability to communicate with people who speak that language. In today's globalized world, being able to speak more than one language is becoming increasingly important. If you can speak a second language, you can communicate with people from other countries and cultures, and you can expand your business, career, and social opportunities. For example, if you work for a multinational company, being able to speak the language of your overseas colleagues or clients can help you build better relationships and close more deals. If you travel to a foreign country, being able to speak the local language can make your experience much more enjoyable and enriching, as you will be able to understand and interact with the local culture more easily. In addition to the practical benefits of learning a new language, there are also cognitive benefits. Learning a new language requires a lot of mental effort, as you have to memorize new vocabulary, grammar rules, and pronunciation. This mental exercise has been shown to have a positive effect on cognitive abilities such as memory, attention, and problem-solving. In fact, studies have shown that people who are bilingual or multilingual are better at tasks that require mental flexibility and executive control, such as switching between tasks, inhibiting irrelevant information, and focusing attention. Learning a new language can also delay the onset of dementia and other age-related cognitive decline, as the constant mental exercise helps keep the brain active and healthy.
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp4$
```